TLE / ICT 9
THIRD QUARTER

LESSON 3

CSS

Formatting Layout with CSS

Box Model

To understand layout using CSS, you have to use the box model where each element on a page exists in its own box wherein each box aligns with other boxes on the page.

The box model consists of margins, borders, padding and the actual content.

CSS allows you to:

- Control the dimension of the box using the height and width properties.
- Define boxes for your content using the block level tags like <p>, <h1>, <table> and the generic block level tag which is placed around other tags to organize the contents' layout.

Heading

Text

Text

Image

Image

Image

Element Positioning

Position property allows placing an object or element precisely on a page. There are four position schemes that you can use: *static*, *relative*, *absolute* and *fixed*.

The position can also follow or inherit the specified position of the body.

Default
position

Relative
top: 100px;
left:100px

Absolute
top: 100px;
left: 100px

Fixed
top: 100px;
left: 300px

CSS POSITION PROPERTY

Property	Value	Description
position:	static	The default or normal position of an object if one would not specify a position.
position:	relative	The position of the box relative to where the default or normal position of an object.
position:	absolute	The position is defined by the values where the position of the element would be from the specified reference point. Ex. from the left, from the top
position:	fixed	The same as the absolute position, the only difference is that the object is “fixed” and will not move with the scroll action of the page.

Offset

- Offset property allows you to set up the distance pf the specified element from another element. This will allow you to place contents of your page in a precise position within the browser window.
- Offset property works together with the position property.
- Before you can use an offset property, you have to set up the position property first.

CSS OFFSET PROPERTY

Property	Value	Description
top:	<value>	Sets how far the box's top content is offset from another element which may be the edge of the page.
right:	<value>	Sets how far the box's right content is offset from another element.
bottom:	<value>	Sets how far the box's bottom content is offset from another element.
left:	<value>	Sets how far the box's left content is offset from another element.

Using the position and offset property, on can actually create a table purely in CSS even without using HTML table set up commands. Below is the example:

```
div#left {width: 270px;
          height: 600px;
          position: absolute;
          top: 120px;
          left: 20px;}
div#right{width: 400px;
          height: 700px;
          position: absolute;
          top: 125px;
          left: 25px;}
```

Float

Float property will allow you to control where a specified block element will be located or placed on the page and where the other elements will be positioned in relation to the said block. When using the float property, it is important to set the width of the block for better control of the float.

```
div#left {width: 300px;
          float: left}
div#right{width: 300px;}
```

Email is the transmission of electronic messages over the internet.
Email is the transmission of electronic messages over the internet.
Email is the transmission of electronic messages over the internet.
Email is the transmission of electronic messages over the internet.

float:right

float:right

Email is the transmission of electronic messages over the internet. Email is the transmission of electronic messages over the internet. Email is the transmission of electronic messages over the internet. Email is the transmission of electronic messages over the internet. Email is the transmission of electronic messages over the internet.

float:right

Email is the transmission of electronic messages over the internet. Email is the transmission of electronic messages over the internet. Email is the transmission of electronic messages over the internet. Email is the transmission of electronic messages over the internet.

CSS FLOAT PROPERTY

Property	Value	Description
float:	left	The specified block element “floats” to the left and all other elements near it will float away from the block.
float:	right	The specified block element “floats” to the right and all other elements near it will float away from the block.
float:	none	This option, specified block element “floats” to the left and all other elements near it will float away from the block.

Wrapping Text around an Element

We can wrap text around an element by using the *float* property in CSS.

Untitled | Float.txt

```
<!DOCTYPE html>
<html>
<head><title>Float</title></head>
<style type="text/css">
body {font-family:Arial;}
h1 {text-align:center; color:#333399;}
h2 {text-align:center; color:violet;}
div {padding:10px; margin:5px;}
img {margin:10px; float:left;}
</style>
<body>
<h1>The Fox and the Grapes</h1>
<div>
<img src="fox.png">
<h2>Paragraph 1</h2>
<p><font size="4">One day the fox was very hungry.  He searched everywhere for food.
At last, the fox went in a vineyard.  He saw some bunches of grapes hanging from the vines.
</font></p><br><br><br>
<div>
<img src="foxgrapes.png">
<h2>Paragraph 2</h2>
<p><font size="4">He tried to reach them but they were too high.  He jumped again and again
but could not reach them.</font></p><br><br><br>
<div>
<img src="foxtired.jpeg">
<h2>Paragraph 3</h2>
<p><font size="4">At last he got tired and said, "these grapes are sour and I don't want them."
Then the fox ran away.</font></p>
</body>
</html>
```

Background Image

We can specify a background image on any element by setting the location as specified in the url (“location”).

b_image.txt image.css

```html
<!DOCTYPE html>
<html>
<head><title>Background Image</title>
<link rel="stylesheet" href="image.css" type="text/css">
</head>
<body>
<h1>The Fox and the Grapes</h1>
One day the fox was very hungry.<br>
He searched everywhere for food.<br>
At last, the fox went in a vineyard.<br>
He saw some bunches of grapes hanging from the vines.
</p><br><br><br>
<p>He tried to reach them but they were too high.<br>
He jumped again and again<br>
but could not reach them.</p><br><br><br>
<p>At last he got tired and said,<br>
“these grapes are sour and I don’t want them.”<br>
Then the fox ran away.</p>
</body>
</html>
```

b_image.txt image.css

```css
body{
background-image:url(https://www.behance.net/gallery/25608867/The-Fox-and-the-Grapes);
    }
```

Background Repeat

- Background image can be repeated by using the *background-repeat* property in CSS.
- You can control how an image background should repeat in order to cover the whole area of a web page or the desired area.

CSS BACKGROUND REPEAT PROPERTY

Property	Value	Description
background -repeat:	repeat	Repeats background (default)
	repeat-x	Repeats background horizontally
	repeat-y	Repeats background vertically
	no repeat	Do not repeat the background
	inherit	Follow the parent’s repeat setting

Background Position

Background can be located where the image will be positioned on the web page by specifying a certain value or keyword location.

CSS BACKGROUND POSITION PROPERTY

Property	Value	Description
background -position:	x y for example: 100px 50px or 20% 10%	Position is specified by the value of percentage.
	for example: top left or center or bottom right or top right or bottom center	Keyword is used to set the position on the page.

Background Attachment

This property allows you to control how the background scrolls with the web page. There are three types: scroll, inherit and fixed.

CSS BACKGROUND POSITION PROPERTY

Property	Value	Description
background-attachment:	scroll	Background scrolls with document (default setting)
	inherit	Follow parent's attachment.
	fixed	Background is steady and does not follow scroll action.

Notes

You can also use a text reverse in your heading using CSS.

h1{color:#ffffff; background-color:blue; padding:.50em}

Setting Margins

- Margins can be set in your webpage by using the *margin* property in CSS.
- You can use the values for the *top, bottom, left and right* margins around an element.

CSS MARGIN PROPERTY

Property	Value	Example
margin-top:	<value>	margin-top: 10px
margin-bottom:	<value>	margin-bottom: 10px
margin-left:	<value>	margin-left: 5px
margin-right:	<value>	margin-right: 5px
margin:	<top><right> <bottom><left>	margin: 15px